SAM EDWARDS

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SOFTWARE ENGINEER / DEVELOPER

Computer Scientist with proven experience in data engineering, machine learning, and software development. Enjoys collaborating with others to design and deliver complex scalable solutions. Known for strong analytical, verbal, and written communication skills. Industries: Technology & Software, Healthcare & BioTech, Robotics & Autonomous Systems

Full Stack Development	Embedded Systems	Verification Testing
Version Control (Git)	Data Pipelines	Machine Learning
Continuous Integration (DevOps)	User Interface Design (UI)	Containerization
Artificial Intelligence (AI)	User Experience (UX)	LLM Integration

TECHNICAL KNOWLEDGE

Languages: Python**,** C#, C++, Java, HTML/CSS, YAML, JavaScript, SQL, Bash
Frameworks & Technologies: REST APIs, Git (GitHub, GitLab), Flask, Docker, React
Databases & Data Platforms: MongoDB, BigQuery
IDEs: Google Colab, Visual Studio Code, Eclipse, PyCharm
Project Applications: Microsoft Word, Excel, PowerPoint, Zoom, Google Suite, ChatGPT, Gemini, Claude Code
Practices: Agile/Scrum, CI/CD Pipelines, Automation, System Integration, Unit Testing
Creative Digital Systems: Cubase, FL Studio, Unity, Unreal Engine, Audacity

EDUCATION AND RESEARCH

Bachelor of Arts - Computer Science Fall 2022 – Spring 2026
Saint John's University, Collegeville, Minnesota
Honors: Magna Cum Laude, Dean's List Spring 2023, Spring 2025, Fall 2025, Spring 2026
Relevant Coursework: Abstraction, Data Structures, and Large Software Systems; Software Development; Discrete Computational Structures; Computer Organization; Artificial Intelligence; Machine Learning; Video Game Design and Development; Theoretical Foundations of Computing; Algorithms & Concurrency; Database Systems

Capstone Project: Bipedal Humanoid Continuous Control Locomotion Fall 2025

- Designed and developed a continuous control reinforcement learning environment in Unity to train a digital bipedal humanoid agent using the ML-Agents Toolkit and Proximal Policy Optimization (PPO).
- Engineered a high-dimensional, multi-component reward function balancing static equilibrium, forward locomotion, and energy efficiency, solving for complex biomechanical articulation.
- Scaled training efficiency by parallelizing 64 concurrent agent instances, converging on stable physical policies.

Artificial Intelligence Project: Autonomous Racing Car Fall 2024

- Led a student team to design, build, and program a physical autonomous RC car utilizing the Donkey Car platform, Raspberry Pi, and camera sensors.
- Engineered computer vision edge-detection and line-following algorithms enabling autonomous track navigation and adaptive speed control on straightaways and curves.
- Integrated hardware components including 3D-printed mounts, PCA boards, and microcontrollers.

Machine Learning Project: Simple Music Genre Classification Using MFCCs Means and Variances Fall 2024

- Designed and implemented a system to classify four music genres (Rock, Electronic, Folk, Classical) using the means and variances of MFCCs (Mel-Frequency Cepstral Coefficients) as primary features.
- Developed and compared three machine learning models (Random Forest, CNN, LSTM) using Python, Tensorflow, and Scikit-learn to evaluate classification accuracy.
- Achieved a 75.5% classification accuracy with an optimized LSTM RNN network, which outperformed both the Random Forest (74.8%) and CNN (73.2%) models.
- Optimized model hyperparameters using Bayesian optimization to maximize performance.
- Validated model stability using 5-fold Stratified K-Fold cross-validation, confirming the LSTM as the most consistent model with a standard deviation of 0.0066.

Research Internship: Breast Cancer Deep Learning / Machine Learning Image Detection Summer 2024
Saint John's University, Collegeville, MN

- Cleaned image datasets to determine standardized size and classification, prevented overfitting using techniques such as image rotation, dropout, and noise.
- Utilized libraries to run and modify individual models; modified and optimized hyperparameters such as number of epochs, learning rate, optimizer used and batch size.
- Achieved an accuracy of ≈97% and an F1 score of ≈94% on over 550k ROI, Mammogram and Histology slide images.
- Collaborated with faculty and team members to achieve project goals.
- Collected, documented, and presented project data and progress maintaining project timeline.

PROFESSIONAL EXPERIENCE

Data Engineer March 2025 – Present
Collective Measures, Minneapolis, MN

- Developed and automated scalable data pipelines using SQL and Google Cloud Services.
- Engineered and optimized complex SQL queries within BigQuery for analysis.
- Used Docker and Google Cloud Services for consistent, reliable deployments of automated reports.
- Created and maintained Dockerfiles to containerize Python applications.
- Managed infrastructure configurations and CI/CD pipeline definitions using YAML and SQL.
- Integrated HTTP compression to reduce payload sizes; Standardized backend caching with global Time-to-Live tiers.
- Streamlined CI/CD workflows by creating a Simple PR Auto Manager tool for GitHub, implementing automatic branch updating, default squash-merging, and dynamic PR queueing.
- Led the technical evaluation for a Google Meridian 2.0 upgrade, mapping out JAX backend migrations and Python 3.11 requirements for Bayesian Marketing Mix Modeling (MMM).
- Designed an evaluation framework for Google Meridian GeoX to conduct geospatial incrementality testing.

A/V Support Technician September 2023 – May 2026
Saint John's University, Collegeville, MN

- Provided technical support for students, staff, and faculty on hardware and software problems including operating system support, application support, and system access support.
- Entered data in Help Desk tracking system and assisted with maintenance.
- Supported clients at live events, filmed and edited video productions, and recorded and mixed audio.
- Created technical and procedural documentation and videos for use by staff.

Electronic Artist, Sound Engineer and Producer 2019 – Present
Sam Edwards Productions

- Created original audio works, including compositions, arrangements, soundscapes, and designs.
- Worked closely with artists, vocalists, and other producers to develop and refine musical ideas.
- Designed and developed unique samples, loops, audio software, and sound libraries.

STUDENT ENGAGEMENT

Wind Ensemble, Symphonic Band, Jazz Band, and Brass Choir; CSB/SJU Fall 2022 – Spring 2026
Saint John's Swimming and Diving; SJU Fall 2022 – Spring 2024